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Institute of Neuroscience and Medicine — Computational Biomedicine (INM-9)
Forschungszentrum Jülich GmbH
52428 Jülich, Germany
Reference: 2025D-126

Re: PhD Position — Simulations and Machine Learning Applied to Proteins (2025D-126)

Dear Hiring Committee,

I am applying for the doctoral position in simulations and machine learning applied to proteins at INM-9 (Ref. 2025D-126). My background sits at the intersection that makes this position attractive: I have developed formal ML frameworks for physics-based simulation surrogates and first-principles computational tools for molecular structure and dynamics, without having run production MD campaigns in GROMACS or AMBER. I am therefore a candidate whose ML and molecular physics depth is established, and for whom the classical MD and enhanced sampling methodology is what the PhD provides — the same way experimental microscopy skills are what a super-resolution physics PhD provides.

Machine learning for molecular simulation: surrogate models and training theory. The most technically demanding part of ML-augmented MD is not training a neural network on a trajectory — it is deciding *what* to train, on *which* data, and guaranteeing that the surrogate is competent outside the training distribution. My Purpose framework addresses this directly. The **Backward Training Principle** proves that a surrogate trained exclusively at the *pure-intent regime* — the fully-resolved, single-objective instance of the learning problem — is ϵ -competent on every constrained sub-regime by feasibility inclusion, not generalisation. A Forward Training Collapse Theorem establishes a provably positive Vapnik–Chervonenkis failure gap for the conventional easy-to-hard approach; a Sample Complexity Theorem gives $(N+1)\times$ sample savings for a cascade of depth N . For protein simulation, the pure-intent regime is the fully-folded, minimum free-energy conformation in the absence of solvent and finite-size artefacts: the regime from which all realistic simulation conditions are feasibility restrictions. Training an ML surrogate there, and projecting competence downward to partial folding states and aqueous conditions, eliminates the need to sample the full configurational space. The homo-veloce system demonstrates the pattern in practice: Gaussian process and graph neural network physics solvers distilled into lightweight 1D CNNs, LSTMs, and Transformers for real-time inference — the same pattern the field uses to replace force evaluations with learned potentials.

First-principles molecular physics: zero-parameter spectroscopy and force fields.

The Borgia framework derives electronic transitions, absorption, emission, and vibrational modes of molecules from bounded phase space geometry alone, with zero empirical parameters. The Triple Equivalence Theorem establishes that oscillatory dynamics, categorical state enumeration, and partition operations are formally identical representations of the same molecular object — meaning a GPU shader computing the partition function is formally equivalent to a physical spectrometer. Validated against NIST for nine elements and six molecules with self-consistency errors below 1.63 %. For protein ML, this matters because it provides a parameter-free baseline for molecular force contributions: instead of fitting empirical parameters to reproduce experimental observables, the observable is derived from geometry. The same framework — Beer-Lambert absorption as a Kirchhoff circuit law (Resonant Stack theorem, transmittance $T = 1/e = 0.3679$ in both representations) — applies to light-triggered protein dynamics and photoresponsive residue behaviour, directly relevant to optogenetics applications in the neuroscience context of INM-9.

Protein state and trajectory analysis. The Shakespear platform provides protein folding, trajectory, catalysis, and dynamics analysis via Kuramoto oscillator synchronization in S-entropy coordinate space. The S-entropy coordinates $(S_k, S_t, S_e) \in [0, 1]^3$ — knowledge entropy (spectral power distribution), temporal entropy (frequency span), and evolution entropy (coherence dimensionality) — provide a continuous, reparameterisation-invariant embedding of protein conformational states. Trajectory completion via the fixed-point operator $T = P \circ B \circ K$ (Kirchhoff propagation, backward Viterbi inference, thermodynamic projection) reconstructs full conformational trajectories from sparse observations, with Banach contraction guaranteeing unique convergence. The Triple Equivalence gives the biological interpretation: every stable protein conformation is a bounded oscillatory system, and the S-entropy coordinates are its complete characterisation at the level of classical statistical mechanics. This is a complementary tool to conventional MD: where MD integrates Newton’s equations forward, the fixed-point operator completes trajectories backward from known endpoint states — applicable to folding pathway reconstruction and transition-state characterisation without forward simulation.

HPC computing and large-scale data. I have practical experience working with large-scale scientific datasets on distributed computing infrastructure. The Lavoisier pipeline processes mass spectrometry datasets exceeding 100 GB using Ray distributed computing, memory-mapped I/O, and HPC-scale parallelism; I have written the associated compute allocation documentation for both local cluster and federated multi-site configurations. The experience of structuring a computational proposal — specifying system sizes, simulation timescales, memory requirements, and expected output volume — transfers directly to the GPU/CPU allocation proposals for JUWELS and the FZ Jülich HPC infrastructure.

Background. I hold an MSc in Computational Biology (Universität Tübingen) and an MSc in Pharmaceutical Biotechnology (HAW Hamburg), and I completed doctoral research in Computational Lipidomics at TUM. I have not run production MD campaigns — I am aware the PhD provides that training — but the molecular physics understanding and the ML methodology I bring are at the level the position requires. I am legally resident in Germany with full work authorisation. English is native; German is conversational. I am prepared to relocate to Jülich.

Yours sincerely,

Kundai Farai Sachikonye